

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                     |
|-----------------------------|-------------------------------------|
| <b>Product Name</b>         | <b><u>1-Dodecanethiol</u></b>       |
| <b>CAS No</b>               | 112-55-0                            |
| <b>Synonyms</b>             | Dodecyl mercaptan; Lauryl mercaptan |
| <b>Molecular Formula</b>    | C <sub>12</sub> H <sub>26</sub> S   |
| <b>Molecular Weight</b>     | 202.4                               |
| <b>Recommended Use</b>      | Laboratory chemicals.               |
| <b>Uses advised against</b> | No Information available            |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>A12741</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number      HSR002503**

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                   |              |
|-----------------------------------|--------------|
| Skin Corrosion/Irritation         | Category 1 C |
| Serious Eye Damage/Eye Irritation | Category 1   |
| Skin Sensitization                | Category 1   |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Chronic aquatic toxicity | Category 1 |
|--------------------------|------------|

Label Elements

Signal Word

Danger

**Hazard Statements**

H314 - Causes severe skin burns and eye damage

H317 - May cause an allergic skin reaction

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P273 - Avoid release to the environment

**Response**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P363 - Wash contaminated clothing before reuse

P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Stench

## Section 3 - Composition and Information on Ingredients

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| 1-Dodecanethiol | 112-55-0 | >95      |

## Section 4 - First Aid Measures

Description of first aid measures**General Advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**New Zealand Emergency Tel.**CHEMTREC®  
09 980 6780 or +64 9 980 6780**Inhalation**

Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove from exposure, lie down. Call a physician

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | immediately. If not breathing, give artificial respiration.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                                                                                                                                                                  |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Clean mouth with water. Call a physician immediately.                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides.

### Decomposition Temperature

350 °C

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

#### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Incompatible Materials**

Strong oxidizing agents. Bases. Metals. Water.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

| Component       | New Zealand WEL | Australia | ACGIH TLV    | The United Kingdom |
|-----------------|-----------------|-----------|--------------|--------------------|
| 1-Dodecanethiol |                 |           | TWA: 0.1 ppm |                    |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber, Nitrile rubber, Neoprene, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                 |                                 |
|------------------------------------------------|---------------------------------|---------------------------------|
| <b>Physical State</b>                          | Liquid                          |                                 |
| <b>Appearance</b>                              | Colorless                       |                                 |
| <b>Odor</b>                                    | Stench                          |                                 |
| <b>Odor Threshold</b>                          | No data available               |                                 |
| <b>pH</b>                                      | Not applicable                  |                                 |
| <b>Melting Point/Range</b>                     | -9 - -7 °C / 15.8 - 19.4 °F     |                                 |
| <b>Softening Point</b>                         | No data available               |                                 |
| <b>Boiling Point/Range</b>                     | 266 - 283 °C / 510.8 - 541.4 °F | @ 760 mmHg                      |
| <b>Flammability (liquid)</b>                   | No data available               |                                 |
| <b>Flammability (solid,gas)</b>                | Not applicable                  | Liquid                          |
| <b>Explosion Limits</b>                        | No data available               |                                 |
| <b>Flash Point</b>                             | 128 °C / 262.4 °F               | <b>Method -</b> CC (closed cup) |
| <b>Autoignition Temperature</b>                | 230 °C / 446 °F                 |                                 |
| <b>Decomposition Temperature</b>               | 350 °C                          |                                 |
| <b>Viscosity</b>                               | No data available               |                                 |
| <b>Water Solubility</b>                        | Insoluble                       |                                 |
| <b>Solubility in other solvents</b>            | No information available        |                                 |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                 |
| <b>Component</b>                               | <b>log Pow</b>                  |                                 |
| 1-Dodecanethiol                                | >6.5                            |                                 |
| <b>Vapor Pressure</b>                          | 1.2 Pa @ 25 °C                  |                                 |
| <b>Density / Specific Gravity</b>              | 0.840                           |                                 |
| <b>Bulk Density</b>                            | Not applicable                  | Liquid                          |
| <b>Vapor Density</b>                           | 6.9                             | (Air = 1.0)                     |
| <b>Particle characteristics</b>                | Not applicable (liquid)         |                                 |

### Other information

|                          |           |
|--------------------------|-----------|
| <b>Molecular Formula</b> | C12 H26 S |
| <b>Molecular Weight</b>  | 202.4     |

## Section 10 - Stability and Reactivity

|                                         |                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                              |
| <b>Stability</b>                        | Air sensitive. Moisture sensitive.                                      |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                |
| <b>Hazardous Reactions</b>              | None under normal processing.                                           |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat.                                     |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Metals, Water.                          |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Sulfur oxides. |

## Section 11 - Toxicological Information

### Acute Effects

#### Information on likely routes of exposure

#### Product Information

|                   |                                                                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | May produce an allergic reaction.                                                                                                                           |
| <b>Eyes</b>       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. Sensitization.                                              |
| <b>Skin</b>       | Avoid contact with skin. Causes burns. Skin Corrosion/Irritation. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. |
| <b>Ingestion</b>  | May cause allergic reaction. May be harmful if swallowed.                                                                                                   |

#### Numerical measures of toxicity

##### (a) acute toxicity;

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Based on available data, the classification criteria are not met |

| Component       | LD50 Oral        | LD50 Dermal       | LC50 Inhalation |
|-----------------|------------------|-------------------|-----------------|
| 1-Dodecanethiol | LD50 >5000 mg/kg | LD50 > 2000 mg/kg |                 |

(b) skin corrosion/irritation; Category 1 C

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Category 1                                                       |

| Component                           | Test method                                       | Test species | Study result  |
|-------------------------------------|---------------------------------------------------|--------------|---------------|
| 1-Dodecanethiol<br>112-55-0 ( >95 ) | OECD Test Guideline 429 Local<br>Lymph Node Assay | mouse        | Sensitization |

**Sensitization** May cause sensitization by skin contact

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

| Component                           | Test method                                                | Test species          | Study result |
|-------------------------------------|------------------------------------------------------------|-----------------------|--------------|
| 1-Dodecanethiol<br>112-55-0 ( >95 ) | OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | in vitro<br>Bacteria  | negative     |
|                                     | OECD Test Guideline 476<br>Gene cell mutation              | in vitro<br>Mammalian | negative     |

(f) **carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) **reproductive toxicity;** Based on available data, the classification criteria are not met

(h) **STOT-single exposure;** Based on available data, the classification criteria are not met

(i) **STOT-repeated exposure;** Based on available data, the classification criteria are not met

**Study result**  
**Target Organs**

NOAEL = 50 mg/kg bw/day  
None known.

(j) **aspiration hazard;** Based on available data, the classification criteria are not met

#### Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Fish                | Water Flea    | Freshwater Algae                      | Microtox       |
|-----------------|--------------------------------|---------------|---------------------------------------|----------------|
| 1-Dodecanethiol | LC50 >100mg/L<br>NOEC =100mg/L | NOEC 0.14mg/L | EC50b < 14.5 µg/L<br>NOEC < 14.5 µg/L | NOEC 36.8 mg/L |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Not readily biodegradable

**Persistence** Persistence is unlikely.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| 1-Dodecanethiol | >6.5    | No data available             |

**Mobility** The product is insoluble and floats on water. Spillage unlikely to penetrate soil. Is not likely

mobile in the environment due its low water solubility.

#### Other adverse effects

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

#### Waste treatment methods

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

## Section 14 - Transport Information

#### NZS 5433:2020

|                                |                          |
|--------------------------------|--------------------------|
| <b>UN-No</b>                   | UN1760                   |
| <b>Proper Shipping Name</b>    | Corrosive liquid, n.o.s. |
| <b>Technical Shipping Name</b> | 1-Dodecanethiol          |
| <b>Hazard Class</b>            | 8                        |
| <b>Packing Group</b>           | III                      |

#### IATA

|                                |                          |
|--------------------------------|--------------------------|
| <b>UN-No</b>                   | UN1760                   |
| <b>Proper Shipping Name</b>    | Corrosive liquid, n.o.s. |
| <b>Technical Shipping Name</b> | 1-Dodecanethiol          |
| <b>Hazard Class</b>            | 8                        |
| <b>Packing Group</b>           | III                      |

#### IMDG/IMO

|                                |                                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| <b>UN-No</b>                   | UN1760                                                                  |
| <b>Proper Shipping Name</b>    | Corrosive liquid, n.o.s.                                                |
| <b>Technical Shipping Name</b> | 1-Dodecanethiol                                                         |
| <b>Hazard Class</b>            | 8                                                                       |
| <b>Packing Group</b>           | III                                                                     |
| <b>Marine Pollutant</b>        | Product is a marine pollutant according to the criteria set by IMDG/IMO |

**Environmental hazards**

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**Transport in bulk according to**

Not applicable, packaged goods



## Annex II of MARPOL 73/78 and the IBC Code

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR002503 |
|-----------------------------|-----------|

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH** Not applicable

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| 1-Dodecanethiol | 112-55-0 | X     | X    | 203-984-1 | -      | -   | KE-12851 | X     | X    |

| Component       | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-----------------|----------|------|-----------------------------------------------|-----|------|-------|------|------|
| 1-Dodecanethiol | 112-55-0 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances

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(Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

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Legend

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Training Advice**

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 15-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet